INVESTIGATING THE EFFECTS OF WING SWEEP AND AREA ON PAPER PLANES

Rishit Madireddy

21.02.2025

INTRODUCTION

From the book, “Understanding Flight”, I’ve learnt about many factors that dictate an aircraft’s performance - 2 of these include wing sweep and wing area. To start, I found one of the most basic objects that fly: a paper plane. Being something I would make often when I was younger, I decided to expand my knowledge on the subject while conducting a practical and computational investigation. From this experiment, I would first like to understand the fundamentals of varying sweep angle and wing area and then locate where pressure and velocity magnitudes will be acting on the aircraft exactly.

HYPOTHESIS

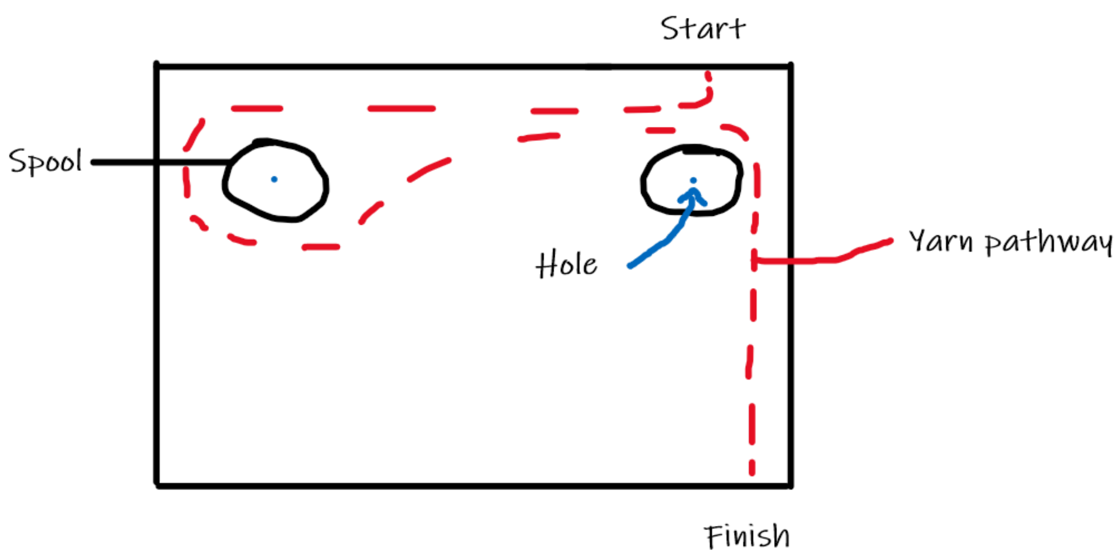
Looking at both sweep angle and wing area, I believe that an increasing wing area and increasing angle between the wings will increase and decrease the distance travelled by the Paperang, respectively. To ensure a fair test, the “style” of the paper aeroplane should be maintained, so I decided to use a “Paperang” design and as for the release mechanism I made a very basic paper plane launcher using the below materials and method.

MATERIALS

- Paper
- Scissors
- Stapler
- Utility knife
- Cardboard box
- 2 skewers
- 1 popsicle stick
- 2 paper clips
- 1 metre of yarn or string
- 2 string spools
- Glue
- A form of weight (e.g. Hole-punch)
- Calculator
- Tape measure
- Stopwatch

METHOD

1. Using a utility knife, cut out a long, rectangular slit at the top of the cardboard box.
2. Bend the paper clips to create a 45° angle on each one.
3. Stick the 2 paper clips onto the edges of the popsicle stick with glue to create a holder for the Paperang.
4. Using a sharp tool (i.e. scissors or utility knife) create 2 holes through the side of the box which are directly below the two ends of the rectangular slit, above.
5. After this, create another hole on the underside of the box - ideally, also directly under one end of the rectangular slit, above.
6. Insert the 2 skewers through each hole on the side - remember to also thread a spool of string on each skewer.
7. Glue one end of the yarn onto the middle of the popsicle stick and thread the yarn as shown in the diagram below.



8. Place the popsicle stick horizontally above the rectangular slit.
9. For better stability, another piece of a popsicle stick can be glued onto the piece of yarn which is underneath.
10. Tie some sort of weight to the string piece which is below the box.
11. Make a Paperang using a stapler, scissors and paper and loosely place the wings of

the Paperang in the “paper clip clamps” while still keeping the popsicle stick and yarn stretched back.

12. Let go of the popsicle stick and use a stopwatch and tape measure to find the distance travelled and time taken.

For a better understanding of the launcher, please refer to the photos uploaded at the end of the report.

Wing area experiment

All of steps 1 to 10 and:

1. Create 3 Paperangs with varying areas.
2. Using the graph paper that you have made the Paperangs from, count the number of whole squares on the Paperang and multiply it by the area of one square.
3. Create a table of results similar to one of the tables below.
4. Stretch the popsicle stick back and make sure that the “paper clip clamps” have a grasp of the wings.
5. Release the popsicle stick.
6. Using a stopwatch and tape measure, record the distance and time taken for the Paperang to fall to the ground.
7. Calculate the average speed by dividing the distance by the time taken.

Wing sweep experiment

All of steps 1 to 10 and:

1. Create 3 Paperangs with varying angles.
2. Using a printed protractor, find the approximate angle between the two wings.
3. Create a table of results similar to one of the tables below.
4. Stretch the popsicle stick back and make sure that the “paper clip clamps” have a grasp of the wings.
5. Release the popsicle stick.
6. Using a stopwatch and tape measure, record the distance and time taken for the Paperang to fall to the ground.
7. Calculate the average speed by dividing the distance by the time taken.

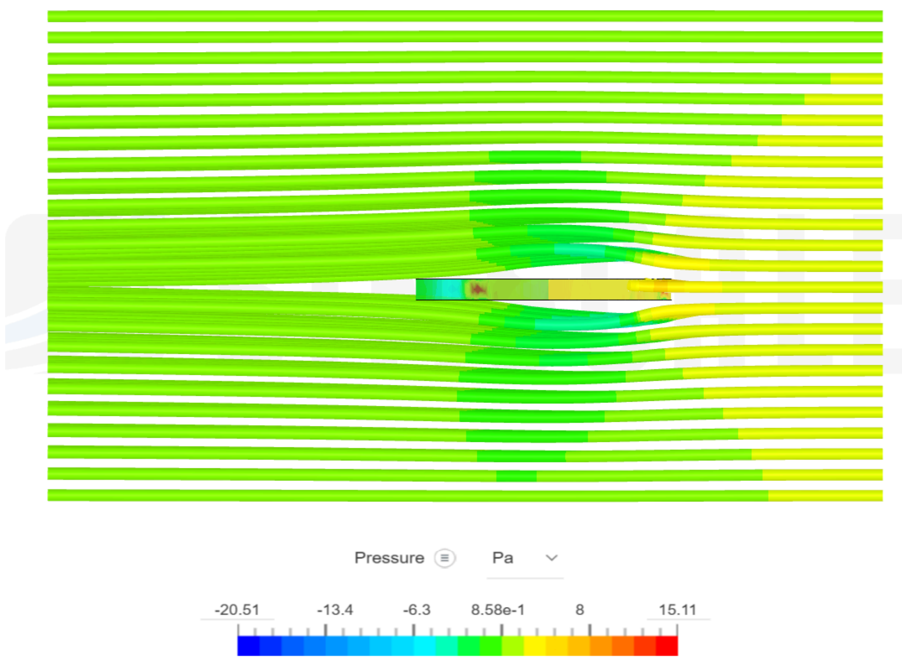
TABLE FOR WING AREA

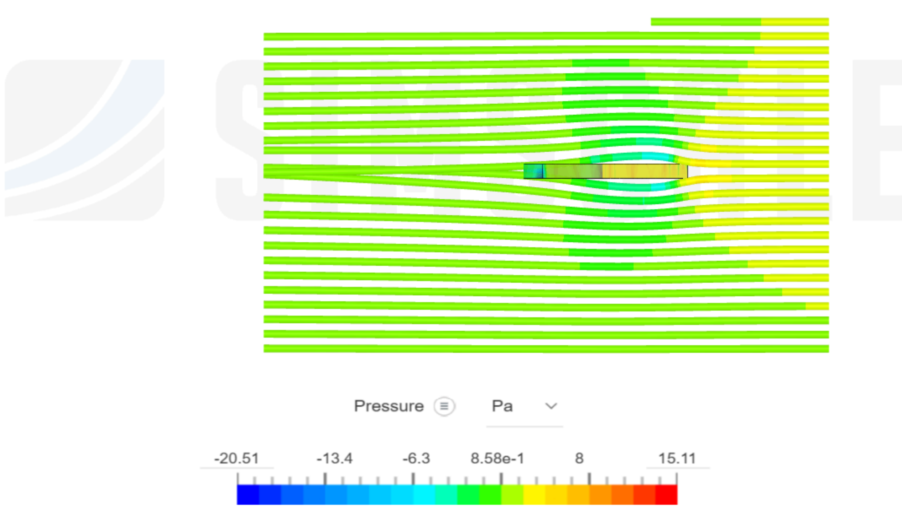
	Area (cm^2)	Distance (m)	Time (s)	Average Speed (m/s)
Paperang 1	84.5	1.49	0.31	4.81
Paperang 2	109	1.27	0.6	2.12
Paperang 3	155	0.97	0.91	1.07

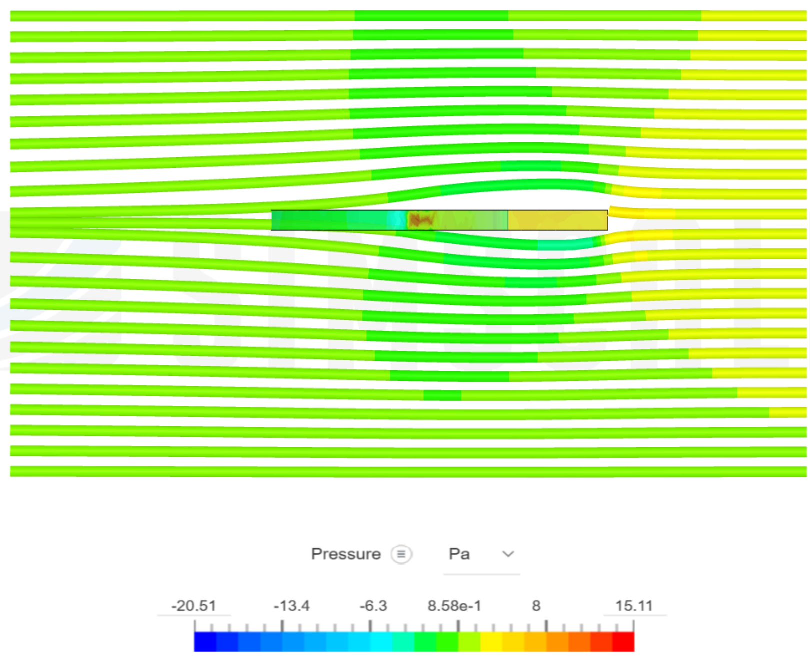
TABLE FOR WING SWEEP

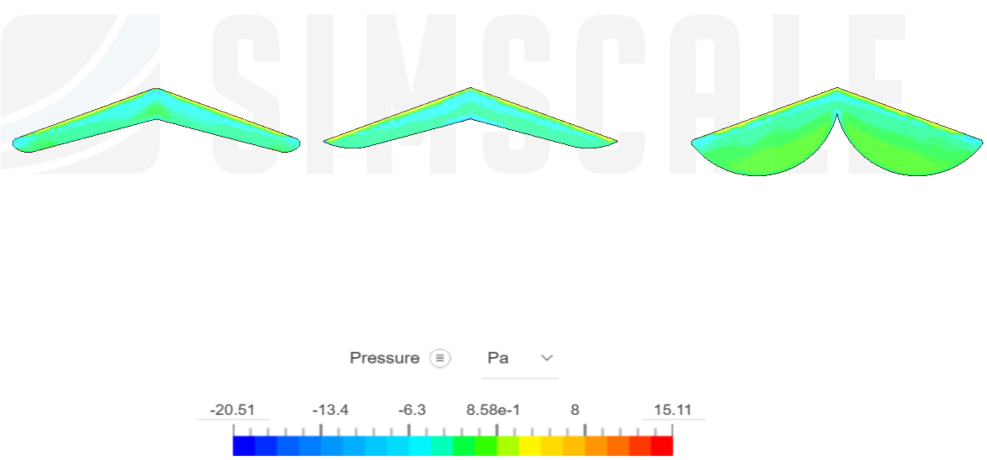
	Angle (°)	Distance (m)	Time (s)	Average Speed (m/s)
Trial 1	162	1.15	1.63	0.71
Trial 2	154	0.18	1.01	0.18
Trial 3	146	0.17	0.73	0.23

SIMSCALE RESULTS






Pressure
Pa
-20.51
-13.4
-6.3
8.58e-1
8
15.11


Pressure
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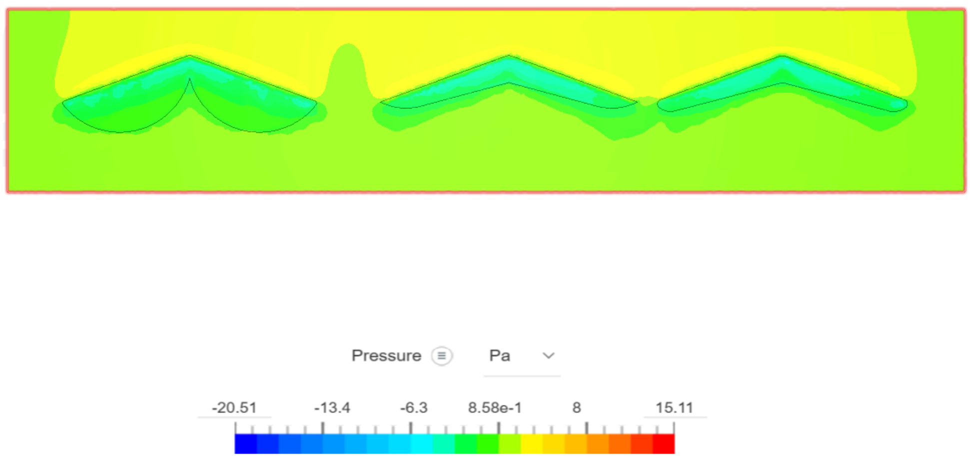
-13.4
-6.3
8.58e-1
8
15.11


Pressure
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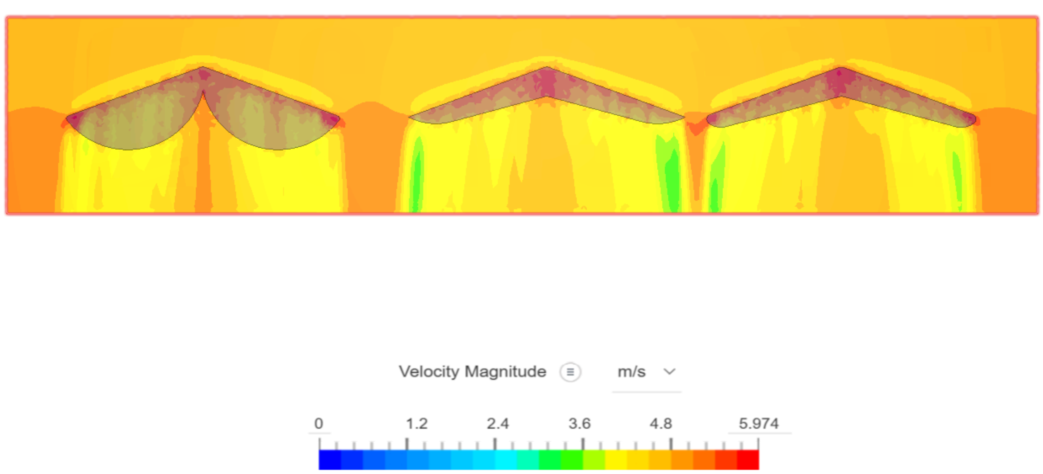
-13.4
-6.3
8.58e-1
8
15.11


Velocity Magnitude
m/s
0
1.2
2.4
3.6
4.8
5.974

ANALYSIS OF RESULTS

From the results in the wing area experiment, I found that the values for area of the wings and the distance travelled are inversely proportional. This means that as wing area increases, the distance travelled decreases. However, in reality, this is not the case as many fighter jets and other aircraft use large-area wings as they significantly contribute to lift and thus time of flight. In this case, I believe the area rule comes into play as there is higher drag towards the middle and tip regions of the Paperang where it would ideally be higher towards the fuselage. From the wing sweep experiment, I have found that the values for the angle between the wings and the distance travelled are directly proportional - which means that as the wing sweep decreases, the distance travelled also decreases. Although, much like wing area, I believe that this factor is more applicable for high speed jets that travel at much larger speeds and thus experience larger forces. To note, for the computational model, I used SimScale along with simplified models to produce the above diagrams showing particle flow, pressure and velocity.

CONCLUSION

Overall, I believe that this experiment was fairly successful. However, there have been a few variables which may have impacted the experiment. These include:

- Inaccurate angle
- Inaccurate counting of squares on wings of Paperang
- Repeated trials not used

These factors can cause anomalies. Taking that into account, I believe that the experiment can be improved by increasing the number of trials. By referring back to my hypothesis, I believe that both statements were correct.

VARIABLES

Independent Variables:

- Wing area
- Angle between wings

Dependent Variables:

- Distance travelled
- Flight time
- Average speed

Control Variables:
Material of Paperang
Launch speed (weight of mass)


Fellowes
WORK BETTER™
LIGHT USE
SHEETS
Sheet Capacity
Cut Type
Particle Size
Safety Shreds
Auto Start/Stop
Run Time